Due Date: To be announced.

Lab Overview

In this lab, your team will extend the heartbeat system from Lab 4 so that either Raspberry Pi can become the leader. The system will contain three voters: Raspberry Pi A, Raspberry Pi B, and a witness program running on a laptop. The witness does not control hardware and cannot become leader. Its only purpose is to vote. A Raspberry Pi must receive at least two of the three votes before it may become leader.

System Components

- Raspberry Pi A is named `pi-a` and has the LED.
- Raspberry Pi B is named `pi-b` and has the buzzer.
- The witness is a Flask program running on a laptop or instructor-provided computer.
- Both Raspberry Pis may become leader. The witness may only vote.

Lab Requirements

Modify the Flask applications from Lab 4 and perform the following tasks:

- Allow both Raspberry Pis to operate as `FOLLOWER`, `CANDIDATE`, or `LEADER`.
- Add an election-term number to heartbeat and voting messages.
- Have the leader send heartbeat messages to the other Raspberry Pi and the witness.
- Start an election when a follower stops receiving valid leader heartbeats.
- When starting an election, have the candidate increase its term and vote for itself.
- Create a `/request-vote` endpoint on each voter.
- Have the candidate request votes from the other Raspberry Pi and the witness.
- Require at least two votes before a Raspberry Pi becomes leader.
- Allow each voter to vote only once during a term.
- Reject messages containing an older term.
- Have a node return to the `FOLLOWER` state when it receives a valid message containing a newer term.
- Save the current term and voting decision so that they are not lost when the application restarts.
- Create a `/status` endpoint that reports the node name, state, term, known leader, and quorum status.
- Record elections, votes, leader changes, and rejected messages in an event log.

Quorum Rule

The system has three voters. A majority requires two votes:

$$\text{majority} = \left\lfloor \frac{3}{2} \right\rfloor + 1 = 2.$$

A Raspberry Pi must not become leader unless it receives at least two votes. The current leader must also receive regular acknowledgments from at least one other voter. If it cannot communicate with a majority for five seconds, it must stop acting as leader and return to the `FOLLOWER` state.

Hardware Behavior

Raspberry Pi A must use the LED as follows:

- LED solid: `pi-a` is the leader and has quorum.
- LED off: `pi-a` is a follower.
- LED blinking slowly: `pi-a` is a candidate.
- LED blinking rapidly: `pi-a` has lost quorum.

Raspberry Pi B must use the buzzer as follows:

- Two short beeps: `pi-b` becomes leader.
- One short chirp every five seconds: `pi-b` is the leader and has quorum.
- Silent: `pi-b` is a follower.
- One short beep: `pi-b` becomes a candidate.
- Three short beeps: `pi-b` loses quorum.

If the LED is solid while the buzzer is producing leader chirps, both Raspberry Pis believe they are leader. This is a system failure that must be investigated.

Testing Requirements

Test the system under the following conditions:

- Start all three voters.
- Start both Raspberry Pis at approximately the same time.
- Stop the follower.
- Stop the leader.
- Restart the former leader.
- Stop the witness.
- Disconnect one Raspberry Pi from the other voters.
- Restore communication to an isolated Raspberry Pi.
- Send the same vote request more than once.
- Send a heartbeat containing an older term.
- Restart a voter after it has already voted.

For each test, record the predicted result, observed result, elected leader, election term, number of votes received, time required to elect a leader, LED and buzzer behavior, and relevant event-log entries.

Safety Requirement

Only one Raspberry Pi may be the authorized leader at a time. A Raspberry Pi without communication with a majority must not continue acting as leader.

- All Flask application and witness source code.
- A LaTeX-generated PDF report.
- A state-transition diagram.
- A diagram showing the two Raspberry Pis and the witness.
- A detailed image of the final hardware setup.
- A table containing the predicted and observed result of each test.
- Relevant event-log entries showing elections and recovery.
- A comparison of the heartbeat system from Lab 4 and the election system from Lab 5.
- A discussion of what happens when a majority is unavailable.

LaTeX Documents – your LaTeX code should be used to generate a PDF which will then be submitted. Be sure that all documents submitted list your name, lab information, date, and class.
Application Code – be sure to submit your source code as indicated above to the course website.